Jalal Abdel Halim

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RESEARCH PROFILE

Ph.D. candidate in Computer Science (University of Toledo, USA; expected 12/2027) working at the intersection of cybersecurity and machine learning. My research covers the adversarial robustness of ML-based intrusion detection in live networks, multimodal (text, image, video) spam and threat detection, and privacy-by-design AI systems, from low-cost edge devices that never share raw camera images to locally hosted large language models that keep sensitive data on-premises. Five publications with IEEE, Springer Nature, De Gruyter Brill, and SciTePress, and extensive hands-on experience building the experimental infrastructure behind this work. Seeking a postdoctoral position in trustworthy and secure AI, network security, or privacy-preserving machine learning.

Keywords: cybersecurity, adversarial machine learning, intrusion detection, trustworthy AI, privacy-preserving AI, large language models, data sovereignty

EDUCATION

Ph.D., Computer Science 2023 – 12/2027 (expected)
University of Toledo Toledo, USA

- Advisor: Prof. Weiqing Sun
- GPA: 3.8/4.0, US scale Focus: cybersecurity, machine learning

M.S., Cyber Security (GPA 3.7/4.0, US scale) 2021 – 05/2023
University of Toledo Toledo, USA

B.S., Information Technology (GPA 3.5/4.0; Dean's List) 2017 – 05/2021
University of Toledo Toledo, USA

RESEARCH EXPERIENCE

Graduate Research Assistant 08/2021 – present
University of Toledo, Department of Electrical Engineering and Computer Science Toledo, USA

- **Adversarial ML for intrusion detection:** Evaluated Random Forest, Decision Tree, and SVM intrusion detectors trained on CICIDS-2017 against black-box Genetic Algorithm and Particle Swarm Optimization attacks. Implemented the live-network validation pipeline on an isolated Proxmox testbed, reconstructing adversarial flows into PCAPs and replaying them with tcpreplay. Showed 100% evasion of tree-based models, and that protocol-driven attacks (FTP, SSH, PortScan) stayed fully functional after perturbation while high-volume floods lost most of their impact (ICISSP 2026).
- **Multimodal spam detection:** Built a real-time, user-trained tool that screens text, images, and video on X/Twitter, Facebook, and Reddit and alerts users before they engage with a post. Benchmarked BERT, DistilBERT, DistilRoBERTa, ResNet-50, MobileNet, and classical baselines on 22,854 tweets and 12,144 images; selected MobileNet for images (92.4% accuracy) and logistic regression for text on the accuracy–latency trade-off (IEEE CSCE 2023; De Gruyter Brill 2025).
- **Privacy-preserving activity monitoring:** Designed a sub-$60 Raspberry Pi and camera system that classifies five daily activities of seniors living alone and shows caregivers an anonymized 3D model on a web dashboard instead of raw images. Collected and labeled 40,724 images and compared six classifiers; EfficientNet-B0 reached 92% accuracy and 93% precision (AIR-RES 2025, Springer CCIS).
- **Privacy-preserving local LLM assistant:** Integrated an on-premises large language model into the course lab platform as a student assistant, keeping all student data on local infrastructure.
- **AI-assisted driver situational awareness:** Co-developed a traffic-light phase predictor that esti-

mates seconds-to-green from dashcam video, using ResNet-18 features from 6-frame sequences; on 3,541 frames from two Toledo intersections, Gaussian Naive Bayes and Random Forest exceeded 91% accuracy with under 3 s mean error. SUMO simulation of a GLOSA-based smart driver (via TraCI) raised average speed by 22% and cut total waiting time by 12% versus baseline traffic at a congested intersection (IEEE MOST 2026).

TEACHING AND OUTREACH

Graduate Teaching Assistant 08/2021 – present
University of Toledo Toledo, USA

- Supported 55+ students across two courses: led hands-on troubleshooting, graded assignments, and explained technical concepts one-on-one.
- Re-engineered the course lab infrastructure, cutting environment setup time from several hours to under 30 minutes.
- Administered concurrent lab environments and user access for all enrolled students, and handled system configuration and hardware and network troubleshooting for the teaching lab.

Workshop Organizer, Linux for Middle School Students
University of Toledo Toledo, USA

- Organized and led a hands-on Linux workshop and competition for 60 middle school students.

PUBLICATIONS

Conference Proceedings

[C1] H. Sun, **J. Abdel Halim**, B. Dwa, E. Chou, W. Sun, "AI-Assisted Driver Situational Awareness for Improved Traffic Flow at High Density Intersections," in *Proc. 2026 IEEE 4th Int. Conf. on Mobility, Operations, Services and Technologies (MOST)*, pp. 86–94, May 2026.

[C2] U. B. Guntupalli, M. Pujari, **J. Abdel Halim**, W. Sun, "Investigating the Influence of Black-Box Adversarial Machine Learning Attacks in Live Network Environments," in *Proc. 12th Int. Conf. on Information Systems Security and Privacy (ICISSP)*, vol. 1, SciTePress, pp. 462–473, 2026. doi:10.5220/0014638900004061

[C3] **J. Abdel Halim**, W. Sun, "A Cost-Effective Tool for Senior Activity Classification and Visualization," in *AI Revolution: Research, Ethics and Society (AIR-RES 2025)*, Communications in Computer and Information Science, vol. 2723, Springer Nature, pp. 267–278, 2026. doi:10.1007/978-3-032-13056-3_20

[C4] **J. Abdel Halim**, W. Sun, "Towards Building a Versatile Tool for Social Media Spam Detection," in *Proc. 2023 Congress in Computer Science, Computer Engineering, & Applied Computing (CSCE)*, IEEE, pp. 2378–2384, Jul. 2023. doi:10.1109/CSCE60160.2023.00385

Book Chapters

[B1] **J. Abdel Halim**, W. Sun, "Empowering Users with an Effective Tool for Social Media Spam Detection," in *Cybersecurity: Cyber Defense, Privacy and Cyber Warfare*, De Gruyter Brill, pp. 333–353, 2025. doi:10.1515/9783111436548-015

PRESENTATIONS

- "Investigating the Influence of Black-Box Adversarial Machine Learning Attacks in Live Network Environments," College of Engineering Design Expo, University of Toledo, 2026.

INDUSTRY EXPERIENCE

Information Technology Intern 05/2022 – 08/2022
O-I Glass Perrysburg, USA

- Built Power BI reports combining data from Rapid7 and ServiceNow to give security and IT staff a consolidated view of system status.
- Wrote automation scripts for data extraction, improving the speed and accuracy of recurring administrative tasks.
- Managed wireless access points and device security using enterprise network-management platforms, and coordinated projects with global colleagues.

SKILLS AND CERTIFICATIONS

- **Machine Learning / AI:** transformer text models (BERT, DistilBERT, DistilRoBERTa), CNN image models (ResNet-50, MobileNetV2, EfficientNet-B0), classical ML (scikit-learn), black-box adversarial attacks (GA, PSO), local LLM deployment, TF-IDF feature extraction
- **Security:** penetration testing, ML-based intrusion detection, CICIDS-2017, CICFlowMeter, tcpreplay, Kali Linux, Fortinet, FortiNAC, Rapid7
- **Infrastructure:** Proxmox VE, VMware ESXi, containerization, Ubuntu Server, Linux system administration, Raspberry Pi
- **Programming & Tools:** Python, Java, Selenium, web data collection (snscrape, Instaloader), Power BI, SUMO and TraCI traffic simulation
- **Certifications:** EXIN DevOps Foundation (2021); ITIL Foundation, AXELOS (2021); Google IT Support Professional Certificate (2021)

REFERENCES

- **Prof. Weiqing Sun** (Ph.D. advisor), Professor, Computer Science and Engineering Technology Program, Department of Engineering Technology, University of Toledo, Toledo, OH. Weiqing.Sun@utoledo.edu
- **Dr. Medha Pujari** (research collaborator), Assistant Professor of Computer Science, Department of Computer Science and Information Technology, Western New England University, Springfield, MA.